

CAPSTONE PROJECT



<https://drive.google.com/file/d/11vXADLGrCrPk8HLMGSDymmCFKqDu9GGp/view>

PROJECT OBJECTIVE



Project Objective

The main objective of this project is to collect, analyze, and visualize web series data to gain meaningful insights. The project starts with **web scraping and data extraction using Jupyter Notebook**, where raw data is collected from online sources (**TVMAZE.COM**). After data collection, the dataset is cleaned and prepared for analysis.

Exploratory Data Analysis (EDA) is performed in Jupyter Notebook to understand data patterns, trends, and relationships between variables such as ratings, popularity, genres, and release years. Based on the cleaned data, a **machine learning model is created** to analyze performance patterns and support predictive insights.

Finally, the processed data is used to build an **interactive Power BI dashboard** that presents key metrics, trends, and top-performing web series. Features like filters and drill-through help users explore the data easily. Overall, this project converts raw data into useful insights that support better decision-making.

DATA COLLECTION METHOD



Data Collection Method

The data for this project was collected using **web scraping through a public API**. The **TVMaze API** was used to extract web series information. Data collection was performed in **Jupyter Notebook** using the **Python requests library**.

The API was accessed page-wise using a for loop to fetch multiple pages of data. For each web series, important details such as title, language, genres, rating, release date, and status were extracted from the JSON response. The collected data was stored in a list and then converted into a Pandas DataFrame.

Finally, the raw dataset was saved as a CSV file for further data cleaning, exploratory data analysis (EDA), machine learning model creation, and dashboard development.

CLEANING STEPS



Cleaning steps

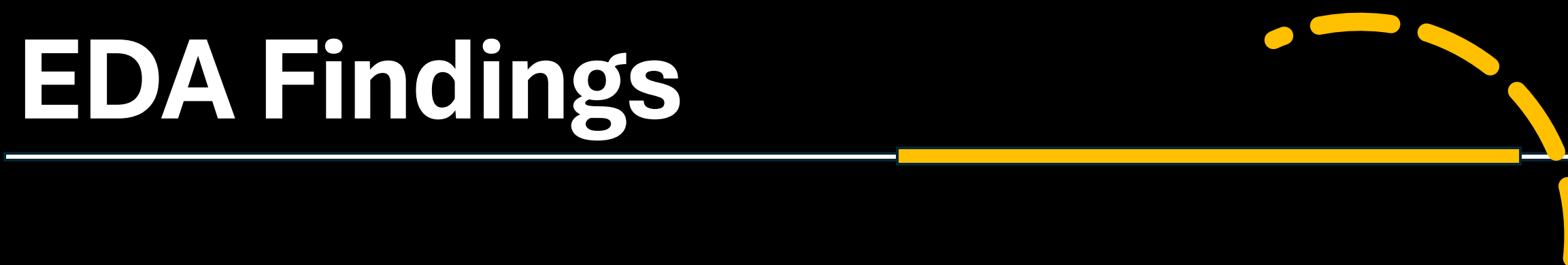
After loading the raw web series dataset in **Jupyter Notebook**, several data cleaning steps were performed to prepare the data for analysis and dashboard creation.

- ❖ The dataset was inspected using **head()** and missing values were identified
- ❖ The **Premiered** column was converted into a **datetime format**, and records with missing premiered dates were removed to maintain time-based accuracy.
- ❖ Missing values in the **Rating** column were filled using the **median rating** to avoid data loss.
- ❖ A new column called **Rating_Percent** was created by converting ratings into percentage format for easier comparison.
- ❖ Missing values in the Genres and **Main_Genre** columns were replaced with **“Unknown”** to handle incomplete data.
- ❖ The dataset was then checked for duplicate records. Although no completely duplicated rows were found, duplicate web series titles existed. These duplicates were removed by keeping only the first occurrence of each title to ensure uniqueness
- ❖ Finally, the cleaned and structured dataset was saved as a new **CSV file**, which was later used in **Power BI for dashboard creation**.

EDA FINDINGS



EDA Findings

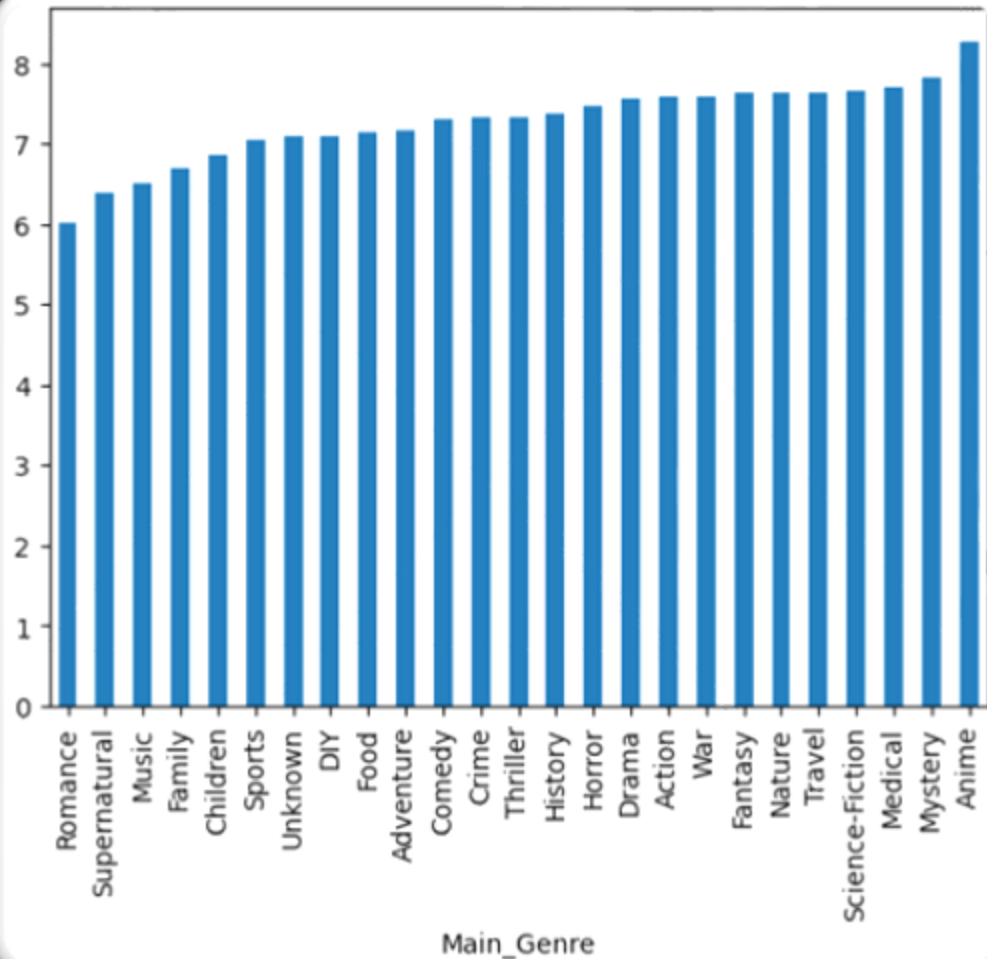
A decorative graphic consisting of a horizontal line that is white on the left and yellow on the right, with a yellow arc curving upwards from the right end of the line.

Popularity Distribution

- Most web series fall under **Medium popularity**.
- Fewer shows are classified as **High popularity**.
- A smaller portion belongs to **Low popularity**.

Conclusion:

Only a limited number of shows achieve high popularity, while most remain moderately popular.

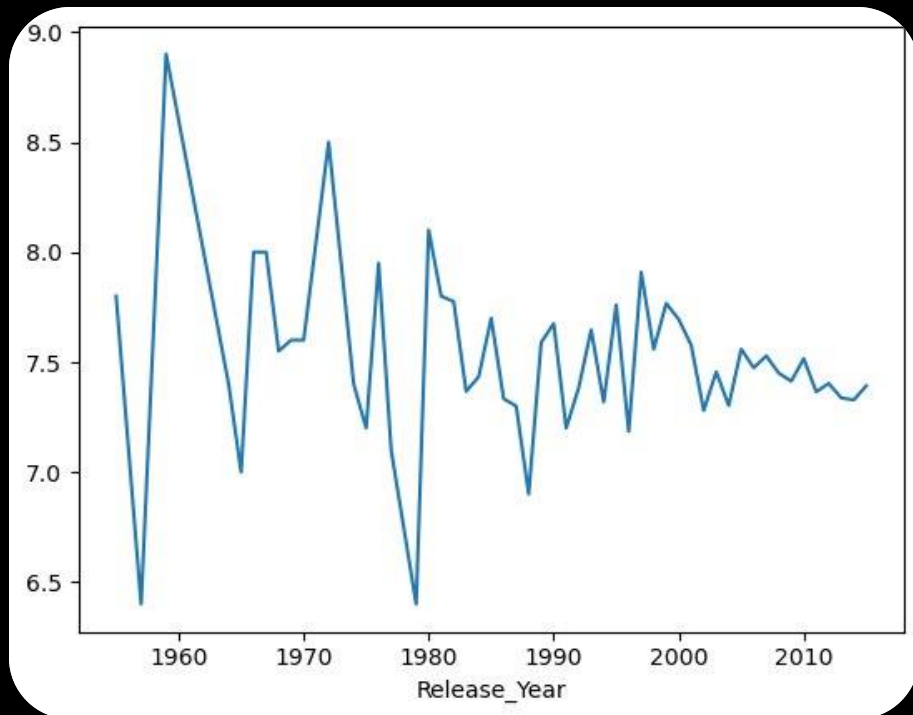


Genre-wise Rating Analysis

- **Anime** has the **highest average rating**.
- **Mystery, Medical, and Science-Fiction** genres also perform well.
- **Romance, Supernatural, and Music** genres have **lower average ratings**.
- **Drama** has good ratings but also has the **largest number of shows**, showing it is **popular but competitive**.

Conclusion:

Some genres consistently perform better, indicating strong audience preference.

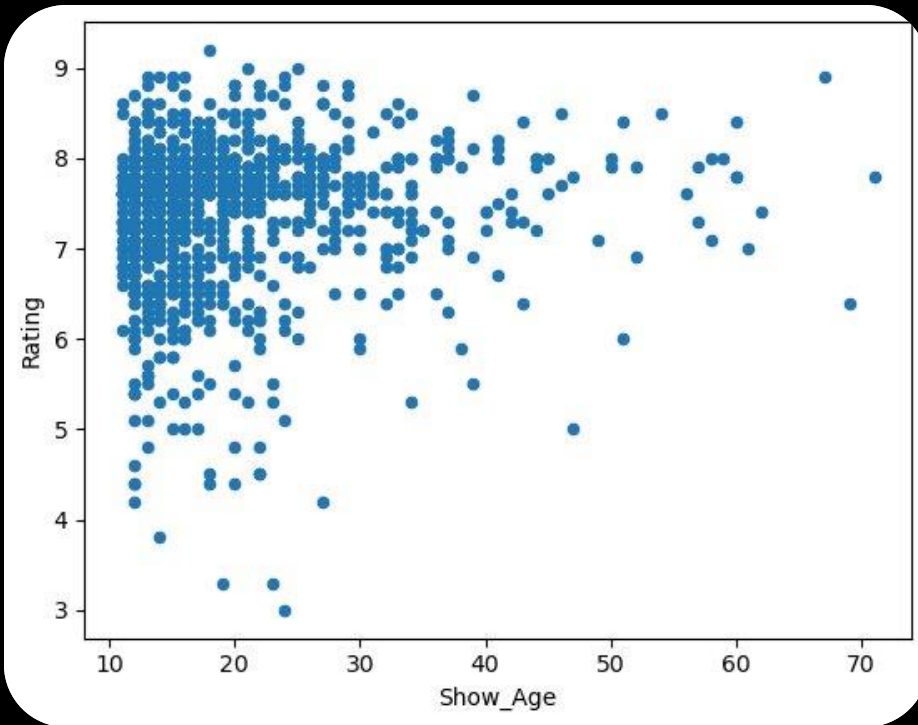


Rating Trend Over Time (Release Year)

- Ratings remain **mostly stable between 7 and 8** across years.
- Older **shows (before 2000)** show **more fluctuation** in ratings.
- **Recent years** show **consistent ratings**, which means content quality has become more stable over time.

Conclusion:

Ratings do not strongly increase or decrease over time, but modern shows are more consistent.

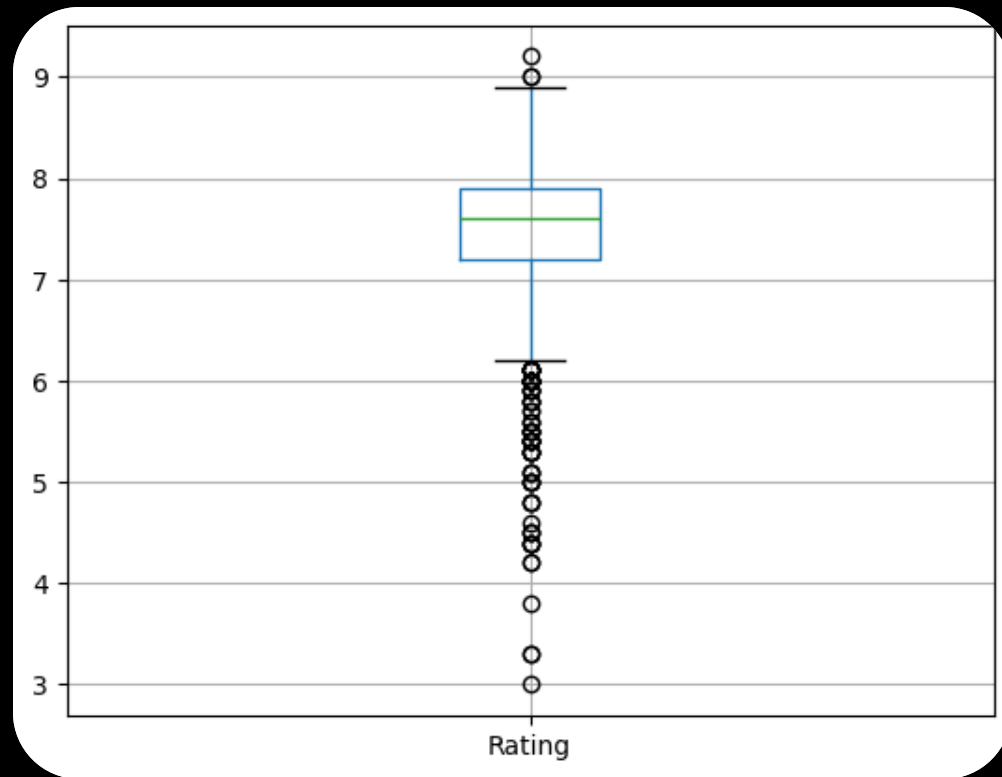


Show Age vs Rating (Scatter Plot)

- Newer shows have **wide rating range**.
- Older shows cluster around **average ratings**.
- There is **no strong relationship** between **show age** and **rating**.

Conclusion:

Show age does not strongly affect ratings.

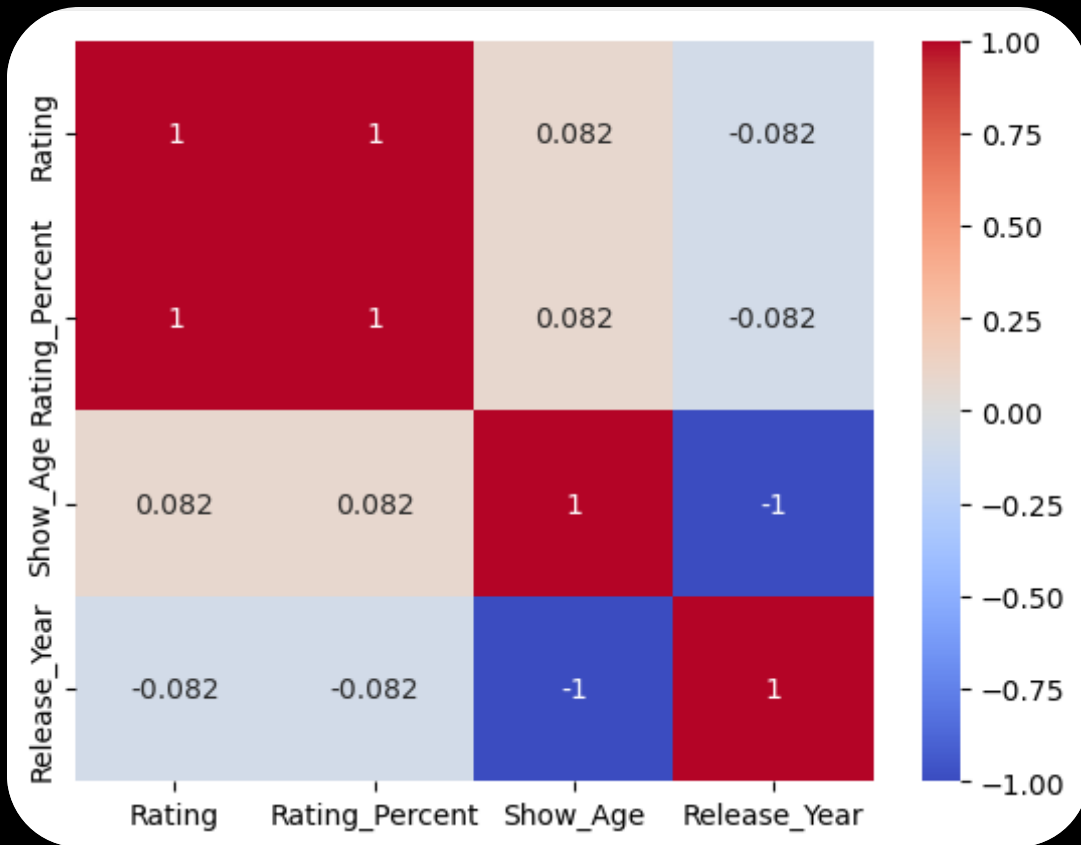


Rating Distribution & Outliers (Boxplot)

- Most ratings lie between **7 and 8**.
- Few shows have **very low or very high ratings**, treated as outliers.
- Median rating is around **7.5**, showing generally good content quality.

Conclusion:

Most web series receive decent ratings, with only a few extreme cases.



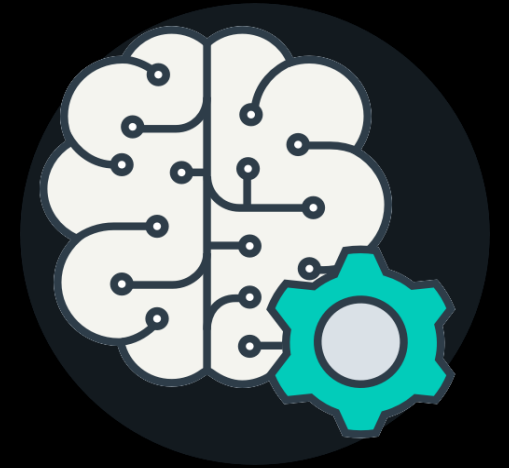
Correlation Analysis (Heatmap)

- Rating and Rating_Percent are **perfectly correlated** (expected).
- Show_Age and Release_Year have a **strong negative correlation**.
- Ratings show **very weak correlation** with age and release year.

Conclusion:

Ratings are mostly independent of time-related features.

MODEL PERFORMANCE



Model performance

The Random Forest classification model achieved an **accuracy of 95.38%**, which shows that the model predicts the popularity of web series very well.

The classification report indicates that the model performs strongly across all classes:

- **High popularity** shows perfect performance with **100% precision and recall**, meaning all high-popularity series were correctly classified.
- **Medium popularity** also shows strong results with **high precision and recall**, indicating reliable predictions for the majority class.
- **Low popularity** has slightly lower precision but still maintains good recall, showing that most low-popularity series were correctly identified.

The confusion matrix further confirms that the model makes **very few misclassifications**, with most predictions falling on the correct class labels.

Overall, the model demonstrates **high accuracy, balanced performance, and good reliability**, making it suitable for popularity prediction based on rating, genre, and time-related features.

KEY BUSINESS INSIGHT



Key business insights

- Most web series fall into **medium popularity**
Only a **small** number of shows reach **high popularity**. This suggests strong competition, where only selected content becomes widely popular.
- **Newer shows** attract better audience engagement
Recently released web series show more consistent and slightly **higher ratings**, indicating a preference for fresh content.
- Machine learning can predict popularity accurately
The **Random Forest model predicts popularity with over 95% accuracy**, proving that features like rating, genre, and release year are strong indicators of success.
- **Genre matters more than release year**
Genres like **Anime, Mystery, and Science-Fiction** receive **higher average ratings**, while genres such as **Romance and Music** perform **lower**. This shows that audience interest depends more on genre than on when the show was released.

- **High popularity** does **not** always mean **high ratings**
Some web series are **highly popular** due to **marketing and visibility**, but their ratings are not always high. This means **popularity and content quality are not the same**.
- **Consistent ratings over time**
Ratings remain mostly **between 7 and 8** across years, showing that viewers expect a certain quality standard from web series.

RECOMMENDATIONS



Recommendations

- Focus on high-performing genres
Content platforms should **invest more** in genres like **Anime, Mystery, Science-Fiction, and Drama**, as they consistently receive higher ratings.
- Improve content quality, not just promotion
Since **high popularity** does **not** always mean **high ratings**, platforms should focus on **strong storytelling and production** quality instead of **relying only on marketing**.
- Support new content strategically
Newer web series tend to **perform better**, so **promoting fresh content** can help **improve audience engagement and satisfaction**.
- Use data to predict success before release
The **machine learning model** can be used to **predict popularity** of upcoming web series, helping platforms make better decisions before investing heavily.
- Leverage dashboards for regular monitoring
Use the **Power BI dashboard** to continuously track **ratings, popularity, and trends**, allowing quick and informed business decisions.

WEB SCRAPING CODE



Web Scrapping Code

```
import requests
import pandas as pd

all_data = []
for page in range(0, 5):
    url = f"https://api.tvmaze.com/shows?page={page}"
    response = requests.get(url)

    shows = response.json()

    for show in shows:
        name = show.get("name")
        language = show.get("language")
        genres = ", ".join(show.get("genres", []))
        rating = show["rating"]["average"] if show["rating"] else None
        premiered = show.get("premiered")
        status = show.get("status")

        all_data.append([
            name, language, genres, rating, premiered, status
        ])

    print(f"Fetch page {page}")
```

```
df = pd.DataFrame(  
    all_data ,  
    columns=["Title", "Language", "Genres",  
            "Rating", "Premiered", "Status"]  
)
```

Df.shape

```
df.to_csv("web_series_raw_1200.csv",  
index=False)
```

```
df = pd.read_csv("web_series_raw_1200.csv")
```


DASHBOARD SCREENSHOTS



TVMAZE

Series Performance Dashboard

7.43
Average Rating

74.30
Avg Rating %

1187
Total Series

1955 - 2015
Release Range

-----Bookmarks-----

- Numbers
- Default
- Charts

Clear all slicers

Apply all slicers

Main_Genre
All

Popularity
All

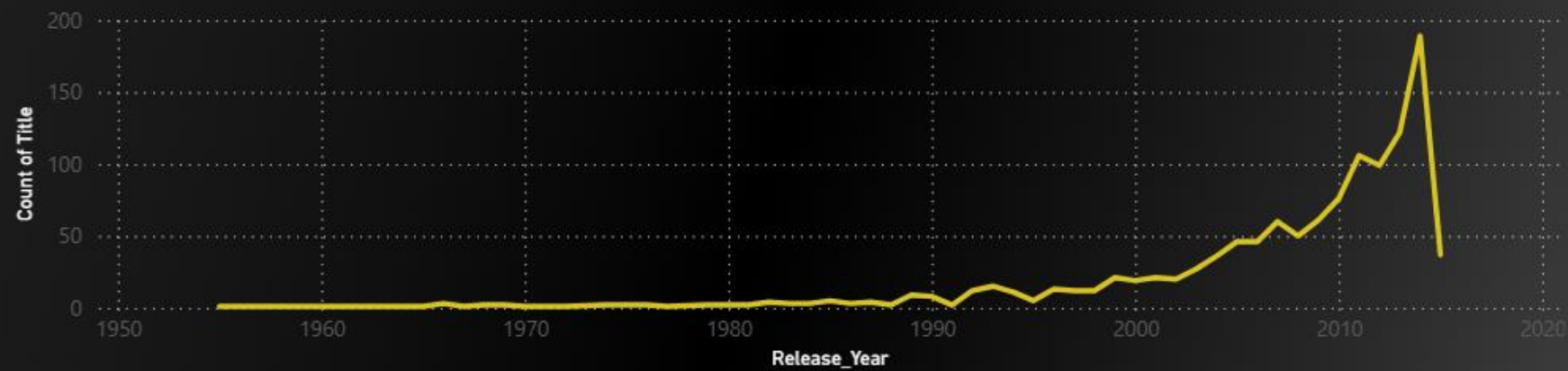
Language
All

Status
All

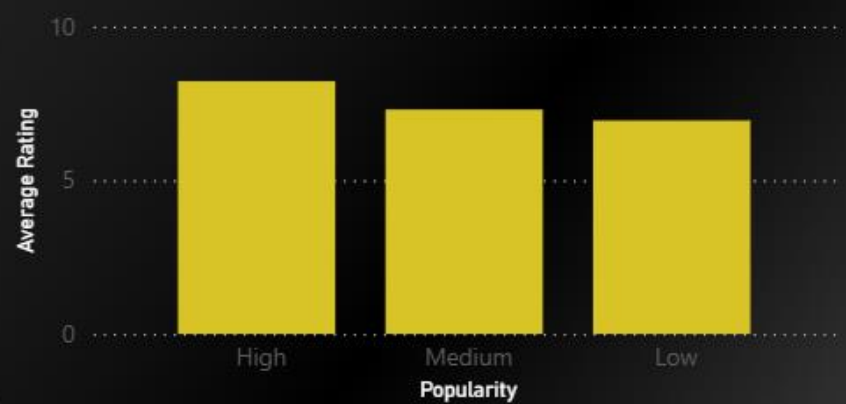
Top Performing Series

Title	Rating
Breaking Bad	9.20
Avatar: The Last Airbender	9.00
Band of Brothers	9.00
Attack on Titan	8.90
Firefly	8.90
Game of Thrones	8.90
Gravity Falls	8.90
Sherlock	8.90
The Twilight Zone	8.90
Death Note	8.80
House	8.80
Person of Interest	8.80
Planet Earth	8.80
Rick and Morty	8.80
Stargate Atlantis	8.80
Stargate SG-1	8.80
The Sopranos	8.80
The Wire	8.80
Battlestar Galactica	8.70
Deadwood	8.70
Downton Abbey	8.70

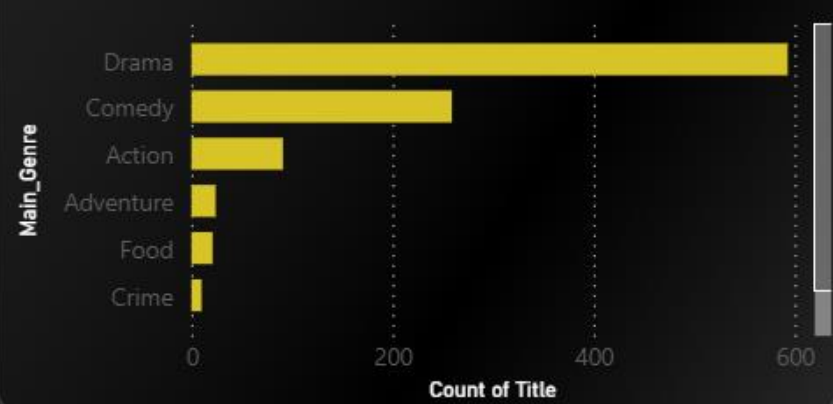
Growth of OTT Over Time



Average Rating by Popularity



Genre Popularity



7.43
Average Rating

74.30
Avg Rating %

1187
Total Series

1955 - 2015
Release Range

AI Insights – What Drives Higher Ratings?

Key influencers Top segments

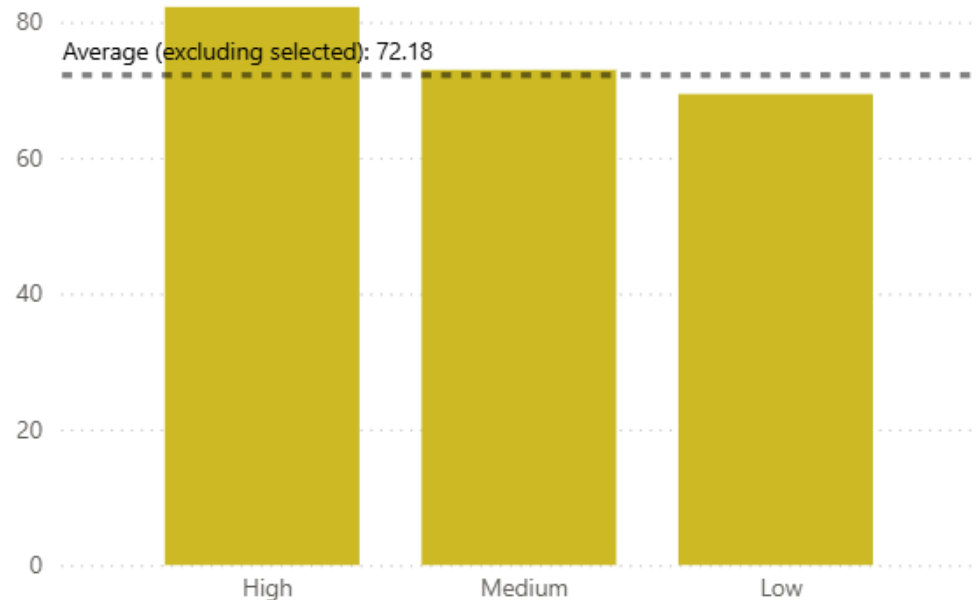
What influences Rating_Percent to Increase ?

When...
...the average of
Rating_Percent increases by

Popularity is High → 10.04

Main_Genre is Drama → 2.95

← Rating_Percent is more likely to increase when Popularity is High than otherwise (on average).



☐ Only show values that are influencers

7.43

Average Rating

74.30

Avg Rating %

1187

Total Series

1955 - 2015

Release Range

AI Insights – What Drives Higher Ratings?

Key influencers Top segments

When is Rating_Percent more likely to be High ?

82.28

82.19

Segment 2

Main_Genre is Drama

Popularity is High

In segment 2, the average Rating_Percent is 82.19. This is 7.87 units higher than the overall average, 74.32.



Segment 2 contains 161 data points (13.6% of the data).



WHAT'S HAPPENING?

- The number of web series increases sharply after 2018
- Drama is the most common genre
- Newer shows generally receive higher ratings
- High popularity does not always mean high ratings
- A small group of shows consistently appear in the Top-rated list

WHY IS THIS HAPPENING?

- The OTT boom after 2018 led to rapid content production
- Drama dominates because it has broader audience appeal
- Audience expectations have increased, so quality matters more than quantity
- Popularity is influenced by marketing, while ratings reflect viewer satisfaction
- Viewers prefer fresh content, leading to higher ratings for recent releases

STORY:-

The dashboard shows that content production increased significantly after 2018, driven by the OTT boom. Drama dominates most, but high popularity does not always guarantee high ratings this indicates that audience satisfaction depends upon quality. Newer shows tend to perform better than old ones, suggesting changing viewer preferences. These insights help us optimize content strategy and investment decisions.



SELECTED SERIES DETAILS FROM DRILL THROUGH

Breaking Bad

First Title

9.20

Sum of Rating

92

Sum of Rating_Percent

English

First Language

High

First Popularity

Ended

First Status